

ALGEBRA QUALIFYING EXAMINATION

Fall 2023

PLEASE WRITE YOUR UO ID NUMBER ON YOUR SOLUTIONS but not your name.

There are three parts to this exam:

- Part I is asking for statements of some definitions/theorems. Three questions, 6 points each, give concise but accurate answers.
- Part II is True/False. Five questions, 8 points each, give a sketch proof or counterexample for full credit.
- Part III is longer problems. Three questions, 14 points each, give concise but clear proofs stating any significant results you are using.

ATTEMPT ALL QUESTIONS

Part I. Briefly state each of the following (6 points each):

1. The definition of an adjoint pair of functors.
2. The Lying Over Theorem (in commutative algebra).
3. The Jacobson Density Theorem.

Part II. True or false? If true provide a brief explanation, if false provide a counterexample (8 points each):

1. There is an isomorphism $D_6 \cong C_2 \times D_3$. Here, C_n is the cyclic group of order n and D_n is the dihedral group of order $2n$.
2. Any field extension E/F of degree 2 is Galois.
3. The $\mathbb{C}$ -algebras $\mathbb{C}[x]/(x^3 - x)$ and $\mathbb{C}[x]/(x^3 + x + 1)$ are isomorphic.
4. There exists a non-zero natural transformation $\text{Id} \rightarrow \bigwedge^2$, where Id is the identity endofunctor of the category $\mathcal{V}ec_{\mathbb{R}}$ of real vector spaces, and $\bigwedge^2 : \mathcal{V}ec_{\mathbb{R}} \rightarrow \mathcal{V}ec_{\mathbb{R}}$ is the functor sending a vector space to its exterior square.
5. Any finite ring R (i.e., a ring of cardinality $|R| < \infty$) with no non-zero nilpotent elements is commutative.

Part III. Give complete solutions to the following (14 points each):

1. Let G be a finite group and $A \leq G$ be an Abelian subgroup.
 - (a) Prove that all irreducible $\mathbb{C}A$ -modules are one-dimensional.
 - (b) Prove that the dimension of an irreducible $\mathbb{C}G$ -module is at most $[G : A]$.

2. This is a question about matrices over principal ideal domains.

- (a) Compute the invariant factors $\delta_1(x)|\delta_2(x)|\delta_3(x)$ of the matrix

$$\begin{pmatrix} x & -1 & 0 \\ -1 & x & -1 \\ 0 & -1 & x \end{pmatrix} \in \text{Mat}_3(\mathbb{Q}[x]).$$

Hence, write down the minimal polynomial of the matrix

$$\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

over the field $\mathbb{Q}$.

- (b) Prove that any matrix $A \in \text{Mat}_n(\mathbb{Z})$ of determinant 6 can be factored as $A = BC$ for $B, C \in \text{Mat}_n(\mathbb{Z})$ with $\det(B) = 2$, $\det(C) = 3$.

3. All algebraic sets in this question are defined over the field $\mathbb{C}$. Suppose that X and Y are *isomorphic* algebraic sets, and that X has precisely two irreducible components X_1 and X_2 .

- (a) Prove that Y also has precisely two irreducible components Y_1 and Y_2 .
 - (b) Prove that the algebraic sets $X_1 \cap X_2$ and $Y_1 \cap Y_2$ are isomorphic.
 - (c) Are the algebraic sets $\mathcal{V}(x^2 - y^2)$ and $\mathcal{V}(x^2 - 1)$ in $\mathbb{A}^2$ isomorphic?